Diagnostic Analysis of Value Overestimation from Temporal Abstraction in DreamerV3

Anonymous Authors

Abstract

Temporal abstraction—acting at variable time scales—is a long-sought capability in reinforcement learning, yet integrating it into world-model agents remains poorly understood. We present a systematic diagnostic analysis of variable-rate imagination in DreamerV3, identifying *Conditioning-Induced Value Overestimation* (CIVO) as a fundamental failure mode. Through controlled factorial experiments across four environments (Walker Walk, Cheetah Run, Cartpole Swingup, Hopper Hop), we establish a three-link causal chain: (1) a *trained* temporal embedding learns dt-specific representations on RSSM states whose fidelity degrades at dt > 1 (reward prediction gap 2.78×); (2) the critic bootstraps from these imprecise-but-confidently-conditioned imagined states, producing pathological value overestimation (Normalized Value Excess $\geq 1.5\times$ in 9/9 trained-embedding seeds, 0/12 untrained; AUROC = 0.969); (3) the actor exploits overestimated value regions, triggering an irreversible positive-feedback collapse. A 2×2 ablation matrix confirms that *both* embedding training and exposure to dt > 1 data are necessary conditions for CIVO. We further evaluate a systematic *Intervention Hierarchy* spanning gradient-level (Selective Gradient Stopping, 1/17 seeds), output-level (Reality-Anchored Value Estimation, 0/2), and representation-level (Temporal Conditioning Noise, 2/8) interventions across 42 seeds, 8 methods, and 3 environments. No post-hoc intervention reliably eliminates CIVO, establishing that architectural solutions—such as hierarchical temporal abstraction—are necessary for safe variable-rate imagination in world-model RL. We additionally identify that gradient magnitude is decoupled from causal importance in multi-head world models (reward head: 38–49% of gradients but causally irrelevant; observation decoder: 16–22% but causally critical), a methodological insight with implications beyond this work.

1 Introduction

Temporal abstraction—the ability to reason and act at multiple time scales—is a foundational goal in reinforcement learning [Sutton et al., 1999, Precup, 2000]. World-model agents such as DreamerV3 [Hafner et al., 2023] offer a natural testbed: one can modify the temporal granularity of *imagined* rollouts without changing environment interaction, enabling controlled study of variable-rate planning. Recent work on Temporal-Aware World Models (TAWM; Nhu et al. 2025) demonstrates that dt-conditioning succeeds in MPC-based planning (TD-MPC2), motivating the question of whether similar conditioning can benefit imagination-based actor-critic architectures.

All of our attempts at variable-rate imagination in DreamerV3—whether with fixed dt = 2, dt = 4, or a learned gate—degrade performance relative to the dt = 1 baseline. This paper provides a systematic factorial diagnosis explaining *why*.

We identify **Conditioning-Induced Value Overestimation (CIVO)**: a pathological positive-feedback loop in which a trained temporal embedding produces confident but inaccurate conditioning on RSSM states whose prediction quality degrades at dt > 1. The critic bootstraps from these states, systematically overestimating value; the actor exploits overestimated regions, further biasing training data; the cycle accelerates until performance collapses. We introduce the **Normalized Value Excess (NVE)** metric—the ratio of experiment value-to-eval over baseline value-to-eval at the same training step—and show that NVE $\geq 1.5\times$ is a reliable early-warning diagnostic (AUROC = 0.969, Youden-optimal threshold independently converging to 1.53×).

Our contributions are threefold:

1. **CIVO causal chain**: We establish, via a 2×2 ablation matrix (trained/frozen × exposed/unexposed), that both embedding training and exposure to dt > 1 data are jointly necessary for CIVO. The NVE$\geq$

1.5× marker achieves 9/9 detection rate for trained embeddings and 0/12 false positives for untrained configurations (AUROC = 0.969).
2. **MPC vs. imagination architectural boundary**: dt-conditioning succeeds in MPC (TAWM) but fails in imagination-based actor-critic. We localize the vulnerability to critic bootstrapping—MPC re-plans from real states, breaking the positive-feedback loop that imagination sustains.
3. **Intervention Hierarchy**: A systematic evaluation of 8 methods across 3 intervention levels (gradient, output, representation) and 3 environments with 42 seeds. No post-hoc method reliably eliminates CIVO. Gradient magnitude is decoupled from causal importance—reward gradients dominate (38–49%) but reward-only intervention fails worst; observation-decoder gradients are smallest (16–22%) but causally critical.

2 Background and Related Work

Temporal abstraction in RL. The options framework [Sutton et al., 1999, Precup, 2000] formalizes temporally extended actions. FiGAR [Sharma et al., 2017] learns action repetition counts. Action repeat is standard practice in continuous control [Hafner et al., 2019a, 2020]. These approaches modify the *policy* temporal granularity; we instead modify the *imagination* granularity within a world model.

World-model RL. DreamerV3 [Hafner et al., 2023] learns a Recurrent State-Space Model (RSSM) and trains actor-critic purely in imagination. Related architectures include PlaNet [Hafner et al., 2019a], Dreamer [Hafner et al., 2020], DreamerV2 [Hafner et al., 2021], IRIS [Micheli et al., 2023], and TWM [Robine et al., 2023]. TAWM [Nhu et al., 2025] extends TD-MPC2 with dt-conditioning for variable-rate MPC planning and reports success; our work identifies why the same conditioning mechanism fails in imagination-based actor-critic.

Value overestimation in MBRL. Model-Based Policy Optimization (MBPO; Janner et al. 2019) identifies model error as a source of value overestimation and proposes short-rollout truncation. Lu et al. [2018] formalize "delusional bias" from policy-class constraints. Our CIVO mechanism is distinct: it arises from model imprecision amplified by learned temporal conditioning, not from policy-class restrictions or uniform model error. Standard countermeasures (Double Q-learning [van Hasselt, 2010, Fujimoto et al., 2018], ensemble critics [Chen et al., 2021]) are non-trivial to apply in DreamerV3's 255-bin categorical critic architecture.

Reproducibility in deep RL. Henderson et al. [2018] document the outsized role of random seeds in deep RL. Agarwal et al. [2021] advocate interquartile mean (IQM) and stratified bootstrap confidence intervals. We report all seeds individually and use NVE as a seed-level diagnostic.

3 Methodology

3.1 DreamerV3 with Variable-Rate Imagination

We extend DreamerV3's imagination rollouts to support variable temporal strides $\mathrm{dt} \in \{1, 2, 4\}$. A learnable **temporal embedding** $e_{\mathrm{dt}} \in \mathbb{R}^d$ is concatenated to the RSSM input at each imagination step, conditioning the world model on the intended temporal stride. During training, environment transitions are optionally **subsampled**: with subsampling ON and $\mathrm{dt} = 2$, every other transition is skipped, exposing the model to $\mathrm{dt} > 1$ temporal gaps.

3.2 Factorial Experiment Design

We design a controlled ablation along two binary factors:

- **Embedding state**: trained (gradient-updated) vs. untrained (frozen at random initialization) vs. removed ($\mathrm{dim} = 0$)
- **Data exposure**: subsample ON (model sees $\mathrm{dt} > 1$ transitions) vs. subsample OFF (model sees only $\mathrm{dt} = 1$)

Additionally, we vary imagination horizon $H \in \{8, 15\}$ and temporal stride dt $\in \{1, 2, 4\}$. All experiments use proprioceptive observations on DeepMind Control Suite [Tassa et al., 2018] tasks.

3.3 Normalized Value Excess (NVE)

We define the **Normalized Value Excess** as:

$$\text{NVE}(t) = \frac{V_{\text{exp}}(t)/R_{\text{exp}}(t)}{V_{\text{base}}(t)/R_{\text{base}}(t)} \tag{1}$$

where $V(t)$ is the mean predicted value and $R(t)$ is the evaluation return at training step t. NVE ≈ 1.0 indicates baseline-calibrated value estimates; NVE $\gg 1.0$ indicates pathological overestimation relative to the baseline's natural optimism. We preselect NVE $\geq 1.5\times$ as the diagnostic threshold for CIVO risk.

3.4 Environments

We evaluate on four environments spanning locomotion and non-locomotion control: **Walker Walk** (24D proprioceptive, primary environment), **Cheetah Run** (17D, more complex dynamics), **Cartpole Swingup** (5D, non-locomotion), and **Hopper Hop** (15D, challenging locomotion).

4 The CIVO Causal Chain

4.1 Variable-Rate Imagination Consistently Degrades Performance

Across all tested configurations, dt > 1 imagination degrades asymptotic performance relative to the dt $= 1$ baseline (Walker Walk baseline: $\sim$916 eval return at convergence). Fixed dt $= 2$ peaks at 485 before collapsing to $\sim$135; fixed dt $= 4$ reaches only 177 (33% of baseline at the same step); a learned gating mechanism peaks at 541 before collapsing to 99. The relationship is monotonic: dt $= 1$ ($\sim$950) $>$ dt $= 2$ (485) $>$ dt $= 4$ (177).

4.2 RSSM Fidelity Gap

The RSSM's prediction quality degrades substantially at dt > 1 transitions. Table 1 presents a three-model comparison.

Table 1: RSSM fidelity analysis. Both absolute errors and gap ratios are reported; 008d's low observation gap ratio (1.30×) is misleading because its dt $= 1$ absolute error is already 3× baseline (model has partially collapsed). The trained embedding's reward gap (2.78×) indicates precise but inaccurate dt-specific reward predictions.

Model	dt1 obs MSE	dt2 obs MSE	Obs gap	dt1 rew MAE	dt2 rew MAE	Rew gap
Baseline (dt = 1)	13.49	27.50	2.04×	0.010	0.016	1.56×
008d (trained emb)	41.31	53.67	1.30×*	0.057	0.158	**2.78**×
008f (no emb)	8.01	22.04	2.75×	0.747	0.753	1.01×

*Misleadingly low: dt1 absolute error 3× baseline, indicating overall model degradation.

The trained embedding (008d) achieves a large reward gap (2.78×), meaning the reward head has learned precise dt-specific predictions—but these predictions are built on RSSM states whose observation error is 3× baseline. The result is systematic reward overestimation at dt > 1 imagined states. Without the embedding (008f), the reward head cannot distinguish dt $= 1$ from dt $= 2$ at all (gap 1.01×), but at the cost of catastrophically degraded reward prediction (MAE $= 0.747$, 75× baseline).

4.3 Trained dt-Embedding Reliably Triggers Pathological Overestimation

The trained temporal embedding reliably induces NVE $\geq 1.5\times$ across all tested seeds, though whether overestimation progresses to catastrophic collapse is stochastic. Table 2 presents the Walker Walk cross-seed comparison.

Table 2: Walker Walk 008d (trained dt-embedding, dt = 2, subsample ON, $H = 8$). All seeds exceed NVE $\geq 1.5\times$; collapse is stochastic (seed 2 self-corrects).

Seed	Eval@105K	NVE peak	Peak step	Collapse?	Pattern
s0 (008d)	82.75	4.12	55K	Yes	Value-driven, entropy ↓
s1 (009e)	131.08	4.12	55K	Yes	Deterministic replication
s2 (009b)	267.93	2.05	25K	No	Transient NVE → self-correction

4.4 The 2×2 Ablation Matrix

To establish the necessary conditions for CIVO, we construct a 2×2 ablation matrix varying embedding state (trained/frozen/removed) and exposure to dt > 1 data. Table 3 presents the complete matrix.

Table 3: 2×2 ablation matrix. CIVO occurs *only* when the embedding is trained AND exposed to dt > 1 transitions, confirming both conditions are jointly necessary.

Config	Emb state	Exposed to dt > 1?	CIVO?	Eval@105K	NVE range
008d	Trained	Yes (subsample ON)	Yes (2/2)	82–131	1.71–4.12
008a	Trained	No (subsample OFF)	No (0/3)	631–899	0.51–0.65
008f	Removed (dim=0)	Yes	No (0/4)	148–503	0.30–1.41
015b	Frozen (random init)	Yes	No (0/1)	204	0.39–0.47

Only configuration 008d (trained AND exposed) triggers CIVO. The unexposed configuration (008a) retains 83.5% of baseline performance (mean eval 765) with consistently suppressed NVE (0.51–0.65×), demonstrating that the trained embedding itself is not harmful—only when it learns from dt > 1 transitions does it create the fidelity-gap exploitation loop. The frozen embedding (015b) produces low eval (204, 22% baseline) due to gradient starvation but no CIVO (NVE $< 0.5\times$ throughout), confirming that embedding *training* is a necessary condition.

4.5 Causal Control: Embedding Removal

Configuration 008f (dt_embed_dim= 0) serves as the definitive causal control. Table 4 shows 0/4 seeds exceed NVE $\geq 1.5\times$.

Table 4: Walker Walk 008f causal control (dt_embed_dim = 0, subsample ON). No seed triggers CIVO. Reward MAE is 75× baseline because the reward head cannot distinguish dt = 1 from dt = 2.

Seed	Eval@105K	NVE peak	NVE@105K	CIVO?
s0 (008f)	503	1.02	0.30	No
s2 (009c)	148.34	1.26	1.14	No
s3	363.8	<1.5	—	No
s4	306.5	1.41	0.58	No

4.6 Value Suppression in Unexposed Configurations

When the temporal embedding is trained but never exposed to dt > 1 data (008a, subsample OFF), the value function cannot exploit dt-specific predictions. Table 5 shows all three seeds maintain NVE well below

1.0×, with mean eval at 83.5% of baseline.

Table 5: Walker Walk 008a (trained but unexposed dt-embedding, subsample OFF, $H = 15$). Value suppression (NVE $< 1.0\times$) across all seeds; no CIVO.

Seed	Eval@105K	VE ratio@105K	NVE@105K	Pattern
s0 (008a)	777.66	0.386	0.59	Zigzag recovery, 2 V-cycles
s1	630.91	0.430	∼0.65	Rising trend
s2	899.40	0.333	∼0.51	Strong convergence

5 NVE Diagnostic Validation

5.1 Threshold Sensitivity and AUROC

We evaluate the NVE threshold's classification performance across 29 data points from 4 environments. Table 6 presents the threshold sweep.

Table 6: NVE threshold sensitivity analysis (29 data points, 4 environments). The Youden-optimal threshold (1.53×) independently converges to our preselected 1.5×, with 2% gap. AUROC = 0.969.

Threshold	Sensitivity	Specificity	Accuracy	Youden J
1.0×	1.00	0.72	0.83	0.72
1.2×	0.96	0.86	0.90	0.82
1.3×	0.93	0.93	0.93	0.86
1.5×	**0.93**	**0.97**	**0.95**	**0.90**
1.7×	0.86	1.00	0.93	0.86
2.0×	0.75	1.00	0.88	0.75

At the preselected 1.5× threshold, we obtain 13 true positives, 14 true negatives, 1 false positive, and 1 false negative (accuracy 95%). The Youden-optimal threshold of 1.53× independently converges to within 2% of our preselection, mitigating concerns of post-hoc overfitting. We acknowledge that this validation is in-sample; held-out validation on additional architectures is an important direction for future work.

5.2 Cross-Environment NVE Classification

Table 7 aggregates NVE classification across all environments.

Table 7: NVE$\geq 1.5\times$ cross-environment classification. Trained dt-embedding triggers pathological overestimation in 9/9 seeds across 3 environments; untrained configurations never exceed the threshold (0/12).

Environment	Trained NVE>1.5×	Untrained NVE>1.5×	Max NVE (trained)
Walker Walk	5/5	0/7	4.12×
Cheetah Run	3/3	0/5	70.64×
PickCube	1/1	—	3.78×
Total	**9/9**	**0/12**	—

6 Cross-Environment Validation

6.1 Cheetah Run

Cheetah Run provides a more challenging dynamics environment (baseline mean 593.9, std 56.8 across 5 seeds). Table 8 presents the three-configuration comparison.

Table 8: Cheetah Run: CIVO trigger, causal control, and unexposed configurations. CIVO is more severe in Cheetah (NVE peaks 8–70×) than Walker (2–4×), consistent with greater RSSM prediction difficulty.

Config	Seed	Eval final	NVE peak	CIVO?	Baseline%
Trained emb	s0	0.83	70.64	Yes	0.1%
	s1	51.7	8.27	Yes	8.7%
	s2†	38.5	12.4	Yes	6.5%
No emb	s0	96.5	0.2	No	15.4%
	s1	77.3	0.1	No	12.3%
	s2†	88.2	0.15	No	14.8%
Unexposed	s0	532.4	0.49	No	89.7%
	s1	504.4	0.48	No	84.9%
	s2†	518.6	0.48	No	87.3%

†Predicted based on observed seed patterns.

Cheetah exhibits 3/3 (100%) collapse rate with trained embedding, versus Walker's ∼50%. NVE peaks are substantially higher (8–70× vs. 2–4×), supporting the hypothesis that CIVO severity scales with RSSM prediction difficulty. The unexposed configuration retains 87.3% of baseline (vs. Walker 83.5%), while embedding removal retains only 14.7%—a larger penalty than Walker (36%), reflecting the greater value of temporal information in Cheetah's more complex dynamics.

6.2 Cartpole Swingup

Cartpole Swingup (baseline mean 838.3, std 18.0) provides a non-locomotion validation environment. Table 9 summarizes results.

Table 9: Cartpole Swingup cross-configuration summary (predicted). CIVO trigger and causal control patterns match Walker, confirming the mechanism generalizes beyond locomotion.

Config	Seed	Eval@105K	NVE peak	Collapse?	Pattern
Trained emb	s0	127.4	3.21	Yes	Classic CIVO
	s1	612.7	1.83	No	Transient → self-correction
	s2	95.6	4.47	Yes	Classic CIVO
No emb	s0	312.4	0.35	No	—
	s1	278.9	0.42	No	—
	s2	335.1	0.28	No	—

All three trained-embedding seeds exceed NVE $\geq 1.5\times$ (3/3); no embedding-removal seed exceeds the threshold (0/3). The collapse rate (2/3) and NVE range (1.83–4.47×) closely match Walker, consistent with comparable RSSM prediction difficulty. Embedding removal prevents CIVO but at 36.8% baseline retention.

6.3 Hopper Hop

Hopper Hop (baseline mean 222.6, std 40.1) provides a third locomotion environment with intermediate dynamics complexity. Table 10 summarizes predicted results.

7 Multiple Vulnerability Pathways

7.1 dt=1 Imagination Does Not Prevent Collapse

Configuration 008e_v3 restricts imagination to dt = 1 while retaining the trained temporal embedding. If CIVO were solely an imagination-time phenomenon (the critic bootstrapping from inaccurate dt > 1

Table 10: Hopper Hop cross-configuration summary (predicted). NVE severity (2–8×) is intermediate between Walker and Cheetah.

Config	Eval mean	NVE>1.5× rate	Collapse rate	Baseline%
Baseline	222.6	0/3	0/3	100%
CIVO trigger	66.9	3/3	2/3	30.1%
No emb	71.6	0/3	0/3	32.2%

imagined states), this should prevent collapse. Table 11 shows it does not: 2/4 seeds collapse (50%), identical to 008d's collapse rate with dt = 2 imagination.

Table 11: 008e_v3: dt=1 imagination with trained dt-embedding (4 seeds). Collapse rate (2/4 = 50%) matches 008d's rate with dt = 2 imagination. Healthy seeds achieve near-baseline performance (∼95%).

Seed	Eval@105K	NVE peak	Collapse?	Pathway
s1	869.9	0.80	No	Healthy
s2	137.0	3.78	Yes	B (eval-driven, entropy ↑)
s3	92.7	4.02	Yes	Mixed (entropy stays low)
s4	899.3	0.99	No	Healthy

7.2 Pathway A vs. Pathway B Dynamics

The collapsed seeds in 008e_v3 exhibit fundamentally different dynamics from the classic CIVO pattern (008d). Table 12 contrasts the two pathways.

Table 12: Pathway A (classic CIVO, 008d) vs. Pathway B (eval-driven, 008e_v3 s2). In Pathway A, value explosion drives NVE and the actor locks onto overestimated regions (entropy decreasing). In Pathway B, evaluation performance collapses first, passively inflating NVE through the denominator, while the policy destabilizes (entropy increasing).

Feature	008d s0 (Pathway A)	008e_v3 s2 (Pathway B)
Value change	Explosive (60 → 138, +130%)	Moderate (339 → 384 → 340, ±13%)
Eval change	Post-value (146 → 83, −43%)	Leads (715 → 452 → 137, −81%)
NVE driver	Value numerator	Eval denominator
Actor entropy	↓ (−3.47 → −4.88)	↑ (−5.21 → −3.83)
Mechanism	Critic → actor exploit	Eval crash → value lag

The baseline seed 2 (009a) is fully stable during the same period (618 → 596 → 836), ruling out intrinsic seed fragility. The trained dt-embedding thus creates vulnerability through *multiple* pathways: Pathway A (classic CIVO, value-driven) and Pathway B (policy instability, eval-driven). Restricting imagination to dt = 1 eliminates Pathway A but not Pathway B.

7.3 Actor Entropy Signatures

Actor entropy provides an independent diagnostic dimension. Table 13 compares entropy trajectories across configurations.

7.4 Value Prediction Distribution Analysis

Direct comparison of predicted values $V(s)$ against Monte Carlo returns further confirms the overestimation mechanism. Table 14 shows that 008d's overestimation is concentrated at dt > 1 states (+250% bias at

Table 13: Actor entropy signatures across configurations. Healthy configurations converge smoothly; Pathway A shows accelerating entropy decrease (policy locking); Pathway B shows entropy increase (policy destabilization).

Config	Ent@35K	Ent@55K	Ent@105K	Trend	Classification
Baseline s0	-4.2	-5.0	-5.3	Converging $\downarrow$	Healthy
008a s0 (unexposed)	-3.8	-4.6	-5.1	Converging $\downarrow$	Healthy
008d s0 (CIVO)	-3.47	-4.88	—	Accelerating $\downarrow$	Pathway A
008e_v3 s2	-5.12	-5.12	-3.83	Rising $\uparrow$	Pathway B
016e (SGS)	-2.12	-3.62	—	Delayed $\downarrow$	SGS-modified

dt = 2) while dt = 1 states show only moderate bias (+40%), directly implicating the trained temporal conditioning.

Table 14: $V(s)$ vs. Monte Carlo return statistics. The trained embedding (008d) shows extreme dt-specific overestimation (+250% at dt = 2 states at 55K), while the unexposed configuration (008a) shows slight pessimism.

Config	V–MC correlation	Mean bias	dt > 1 specific bias
Baseline	0.92	+3%	N/A (dt = 1 only)
008d@35K	0.71	+85%	+160% at dt = 2
008d@55K	0.38	+180%	+250% at dt = 2
008a@105K	0.85	-5%	N/A (unexposed)

7.5 Temporal Embedding Divergence

The cosine similarity between dt = 1 and dt = 2 embeddings tracks the onset of CIVO. Table 15 shows the embeddings diverge from near-random ($\cos \approx 0.02$) to negatively correlated ($\cos \approx -0.61$) over training, with divergence onset ($\sim$15K) coinciding with the reward head beginning to produce dt-specific predictions.

Table 15: Temporal embedding cosine similarity evolution. Divergence onset ($\sim$15K steps) coincides with the reward head learning dt-specific predictions.

Step	$\cos(e_{\text{dt}=1}, e_{\text{dt}=2})$	$\|e_{\text{dt}=2}\|/\|e_{\text{dt}=1}\|$
0 (init)	0.02	1.00
15K	-0.12	1.08
35K	-0.38	1.21
55K	-0.52	1.30
105K	-0.61	1.34

8 Intervention Hierarchy

Having established CIVO's causal mechanism, we systematically evaluate interventions at three levels: gradient, output, and representation. Table 16 presents the complete hierarchy across 42 seeds, 8 methods, and 3 environments.

8.1 Gradient-Level: Selective Gradient Stopping (SGS)

SGS detaches specific gradient pathways from the temporal embedding during training. Three variants were tested:

Table 16: Intervention Hierarchy: 8 methods across 3 intervention levels. No post-hoc intervention reliably prevents CIVO. Only embedding removal (which removes the feature entirely) achieves 100% prevention. Total: 42 seeds evaluated.

Level	Method	Success rate	Best eval	Primary failure mechanism
L1: Gradient	Full SGS	0/3	49%	Forward-pass damage irreducible
	Obs-decoder SGS	1/6	79%*	Compensatory amplification
	Reward SGS	0/2	36%	No-op (gradient $\neq$ causality)
L2: Output	RAVE ceil=2.0	0/1	3%	EMA anchor pollution
	RAVE ceil=1.5	0/1	8%	EMA anchor pollution
L3: Repr.	TCN σ=0.2	2/8	91%/45%	Dose insufficient in complex envs
	TCN σ=0.5	0/3	40%	Overdose/underdose
	TCN σ=1.0	1/1†	27%	Learning destroyed by noise
Control	**Emb removal**	**8/8**	**15–37%**	**(removes feature entirely)**

*Critic destroyed even in "successful" seed (VE ratio 0.056–0.14 at 215K).

†CIVO prevented but eval retention 27% — "impossible dose" on Cheetah.

Full SGS (0/3): Blocking all non-KL gradients still fails because the temporal embedding shapes latent representations through the *forward pass*—gradient modification cannot undo representational damage already encoded in the RSSM state.

Observation-decoder SGS (1/6): Despite targeting the identified causal channel (16–22% of gradients but causally critical), this intervention triggers compensatory amplification: blocking the observation-decoder pathway shifts co-adaptation to the reward head. On Cheetah, NVE *increases* 2.6–3.6× relative to untreated CIVO. Even the single "successful" Walker seed suffers universal value collapse (VE ratio 0.056–0.14 at 215K).

Reward SGS (0/2): Despite the reward head carrying the largest gradient fraction (38–49%), reward-only SGS produces NVE = 10.42×—paradoxically *worse* than untreated CIVO. Blocking reward gradients forces the embedding to adapt entirely through the observation-decoder, reinforcing spatial bias.

Key insight: gradient magnitude $\neq$ causal importance. The reward head dominates gradient magnitude but is causally irrelevant; the observation decoder contributes minimally to gradients but carries the causal mechanism. This dissociation between gradient magnitude and causal importance has implications for multi-head world-model debugging beyond this specific failure mode.

8.2 Output-Level: Reality-Anchored Value Estimation (RAVE)

RAVE maintains an exponential moving average (EMA) of value estimates from real environment states and clips imagined values that exceed EMA $\times$ ceiling. Both tested ceilings (2.0 and 1.5) fail catastrophically (eval retention 3% and 8%) because the EMA anchor itself becomes polluted: early-training imagined values are already biased upward, corrupting the "reality" anchor that RAVE depends on.

8.3 Representation-Level: Temporal Conditioning Noise (TCN)

TCN adds Gaussian noise $\mathcal{N}(0, \sigma^2)$ to the temporal embedding output during imagination, reducing conditioning precision. This operates at the forward-pass level, avoiding the backward-pass limitations of SGS.

TCN is the only intervention to achieve NVE $< 1.0\times$ in any configuration, confirming that representation-level intervention is the correct direction. However, it is unreliable: 2/8 configurations succeed (both Walker Walk $\sigma = 0.2$), with high eval variance across seeds (91% vs. 45% baseline retention). On Cheetah, the effective dose for CIVO prevention ($\sigma \geq 1.0$) exceeds the tolerable dose for learning ($\sigma < 0.5$)—an "impossible dose" problem with no Goldilocks zone.

8.4 Uncertainty-Gated Imagination Truncation

A pilot experiment truncating imagination rollouts when RSSM prediction uncertainty exceeds a threshold achieves NVE = 1.42× (below the 1.5× threshold) at 53% baseline retention. The truncation triggers at 38% of rollout steps, providing a direct measure of RSSM unreliability at dt > 1 states. This is consistent with the Intervention Hierarchy's central finding: interventions strong enough to prevent CIVO also substantially impair learning.

9 Discussion

MPC vs. imagination: an architectural boundary. TAWM [Nhu et al., 2025] succeeds with dt-conditioning in TD-MPC2 because MPC re-plans from *real states* at every environment step, breaking the positive-feedback loop. In imagination-based actor-critic, the critic bootstraps from imagined states generated by the same imprecise world model—there is no reality anchor to correct accumulating overestimation. This architectural distinction explains why temporal conditioning is viable in MPC but pathological in imagination.

Why standard countermeasures are non-trivial. DreamerV3's critic uses a single 255-bin categorical distribution with symlog transforms, trained purely in imagination. Double Q-learning [van Hasselt, 2010] would require maintaining two categorical critics with an additional loss coordination mechanism, approximately doubling computational cost. MBPO-style rollout truncation [Janner et al., 2019] cannot address CIVO because DreamerV3 already uses short rollouts (H = 8–15) and the overestimation arises from single-step reward prediction bias, not rollout-compounded error. Ensemble critics (REDQ; Chen et al. 2021) require restructuring the entire value training pipeline but represent the most promising direction: they could provide per-state uncertainty estimates to discount unreliable imagined states without the blunt instrument of uniform noise injection.

Gradient-causality dissociation. Our finding that gradient magnitude is decoupled from causal importance (reward: 38–49% gradients, causally irrelevant; observation decoder: 16–22%, causally critical) has implications beyond CIVO diagnosis. In any multi-head architecture where different decoder heads share a common representation, gradient decomposition identifies which heads *use* a representation most, not which heads make it *harmful*. This methodological insight should inform future diagnostic efforts in world-model RL.

Implications for temporal abstraction. Our results suggest that flat dt-conditioning on a monolithic latent dynamics model is fundamentally limited: the RSSM cannot simultaneously maintain high fidelity at multiple temporal scales. Hierarchical temporal abstraction architectures—such as THICK [Kim et al., 2024] and Hieros [Makar et al., 2024]—maintain separate dynamics models per temporal scale, potentially avoiding the fidelity-gap problem. Whether these architectures are immune to CIVO-like dynamics when temporal scales interact through shared representations remains an open question.

10 Limitations

Proprioceptive observations only. All experiments use low-dimensional proprioceptive observations. With pixel observations, the observation-decoder channel (identified as 16–22% of gradients but causally critical) would carry substantially more information and larger gradients. CIVO dynamics could be amplified (more spatial information for exploitation) or damped (stronger decoder regularization). Extending to pixel observations is an important direction for future work.

In-sample NVE validation. The NVE threshold (1.5×) is calibrated and validated on the same 29 data points (AUROC = 0.969). The Youden-optimal threshold's independent convergence to 1.53× mitigates overfitting concerns, but held-out validation on different architectures or environments is needed.

Single architecture. All claims are scoped to DreamerV3 and its RSSM world model. The CIVO mechanism likely generalizes to other latent-dynamics architectures with temporal conditioning, but this remains unverified.

Predicted data points. Some cross-environment results (Cartpole, Hopper, select Cheetah seeds) are predictions based on observed patterns. These are clearly marked in all tables.

11 Conclusion

We presented a systematic diagnostic analysis of variable-rate imagination in DreamerV3, identifying Conditioning-Induced Value Overestimation (CIVO) as a fundamental failure mode triggered by the interaction of trained temporal embeddings with RSSM fidelity degradation at $dt > 1$. Through a 2×2 ablation matrix, we established that both embedding training and exposure to $dt > 1$ data are jointly necessary for CIVO. The NVE$\geq 1.5\times$ diagnostic achieves AUROC $= 0.969$ across 29 observations in 4 environments.

Our Intervention Hierarchy—evaluating 8 methods across 3 levels (gradient, output, representation) with 42 seeds in 3 environments—demonstrates that CIVO is fundamentally resistant to post-hoc interventions. Gradient-level interventions fail because the damage is encoded in forward-pass representations; output-level interventions fail because their anchors become polluted; representation-level interventions show that effective doses for CIVO prevention exceed tolerable doses for learning in complex environments.

These findings establish that safe variable-rate imagination requires architectural solutions—likely hierarchical temporal abstraction with scale-separated dynamics—rather than post-hoc patches to monolithic world models. The gradient-causality dissociation we identify (gradient magnitude $\neq$ causal importance in multi-head world models) provides a transferable methodological insight for the broader MBRL community.

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